SOUTH LAKE HOME EDUCATORS



Lesson 4

Programming Basics III: Direct Hit

Semester 1, Week 4

At a Glance

Lesson 4 objectives & materials

By the end of today, students will:

- Explain what collision/overlap detection is
- Explain trigger/area checks vs. solid physical collision
- Explain why destroying two objects on one hit is safe
- Add a working enemy + collision mechanic to their project

What We're Using Today

- Demo laptop with all 8 engines verified
- Godot, for the live lecture demo
- Engine-specific Student Handouts, one per student
- This lesson's Assets folder (new enemy sprite + hit sound - no download needed)
- Your own [YourName]Game project from Lesson 3 - not a new one

IGDA framework tie-in: Programming - collision/physics systems, event-driven programming (introductory).

Today's Agenda



0:00 - 0:05	Welcome back, Lesson 3 recap, quick share-out
0:05 - 0:25	Collision Detection: How Engines Know Things Touched (live demo)
0:25 - 0:45	Cleanup Revisited: Destroying Two Things at Once, Safely
0:45 - 0:50	Placement: Things You Spawn vs. Things You Place
0:50 - 1:00	Introduce lab exercise, hand out engine-specific handouts
LAB TIME	***— Direct Hit —***
1:00 - 1:50	Add an enemy and collision detection to your Semester 1 project
1:50 - 2:00	Exit check: quick share-out



Welcome Back

Lesson 3 recap and a quick share of any working shooting mechanics.

5 min

Welcome Back

5 min

Quick recap of Lesson 3:

- Instances and cleanup - spawning a bullet that flies up and disappears
- Everyone got a working fire key with a cooldown-capped shooting mechanic
- Today: adding an enemy - your bullets finally have something to hit

Open your own project from last week today, not a new one - everything we build adds on top of the character and bullet you already have.



Collision Detection

How does an engine even know two things touched?

20 min

Collision Detection

20 min

Give each object a simple collision shape (a box or circle, not exact pixel art). Every frame, the engine checks if any two shapes overlap - and fires a signal when they do.

Same underlying question, different names, across every engine:

Godot	`Area2D + area_entered signal`
GameMaker	`Collision event`
Unity	`Collider2D (Is Trigger) + OnTriggerEnter2D()`
Unreal	`Event ActorBeginOverlap`



Discussion: where else in a game does this same idea show up? (Coins, doorways, an attack landing, falling into a hazard.)



Cleanup Revisited

One collision, two things to destroy - and why that's always safe.

20 min

Cleanup Revisited

20 min

One collision now needs to clean up TWO objects: the bullet that hit, and the enemy that got hit.

In every engine we use, calling "destroy this" a second time on something already destroyed is completely harmless - the engine just ignores the repeat.



Design principle: it's the Enemy's own script that reacts to being hit - not the Bullet's. Whoever gets hit is the one responsible for what happens next.



Live demo: fire several bullets quickly, including some that miss. Hits destroy the enemy exactly once - misses keep flying and vanish off-screen exactly like last week.



Placement

Things you spawn while playing vs. things you place ahead of time.

5 min

Placement

5 min

Both are still instances - copies of a template. They just come into existence at different moments:

Spawned at runtime

Doesn't exist until the game creates it while playing.

Last week's bullet - created on each fire-key press.

Placed at design time

Sitting in the scene/room/level from the very start.

Today's enemy - placed once, ahead of time.

You'll see this split all semester: bullets, effects, and pickups get spawned; level layout, a boss, and fixed obstacles get placed.



Today's Exercise & Lab

Direct Hit - the first piece of your capstone's actual combat loop.

10 min

Direct Hit

10 min

A single enemy sits on screen. Shooting it destroys both the enemy and the bullet that hit it - added on top of your existing project.

Level 1 - Minimal

The enemy and the bullet that hits it both disappear on contact.

MVP - a complete finish for today.

Level 2 - Complete

Destroyed cleanly and exactly once - no errors, misses unaffected.

Everything works end to end.

Level 3 - Refined

Sound, a distinct look, and/or clean hit-response structure.

Ready to share.



Heads up: this is the first week needing a real collision/physics component - a Rigidbody2D in Unity, an Area2D shape in Godot, overlap events in Unreal. Check that first if nothing happens on contact.

Lab Time Notes



- Most common bug this week: "nothing happens when I shoot the enemy" - almost always a missing or misconfigured collision component. Check that first.
- Second most common: the bullet destroys things it shouldn't, or the enemy never gets destroyed - usually a missing or mismatched tag/group check.
- Students who finish Level 2 early should be pointed at the Level 3 ideas on their handout - not given new unrelated work.
- At 1:50, call time and do a 2-minute share-out: ask 2-3 students to show what they built, across different engines if possible.



Great Work Today!

Next week: multiple enemies at once - loops that spawn and manage a whole row of them.

Keep polishing your combat mechanic at home if you'd like - try one of the Level 3 ideas if you haven't already.